

Q1 Uptime.c

```
dua_ansari@DESKTOP-4DF8P8: ~$ cd xv6-labs-2025
dua_ansari@DESKTOP-4DF8P8: ~/xv6-labs-2025$ make
make: 'kernel/kernel' is up to date.
dua_ansari@DESKTOP-4DF8P8: ~/xv6-labs-2025$ make qemu
mv -f fs.img fs.img.bk
mkfs/mkfs fs.img README user/findtest.sh user/sixfive.txt user/_cat user/_echo user/_forktest user/_grep user/_init user/_kill user/_ln user/_ls use
r/_mkdir user/_rm user/_sh user/_stressfs user/_usertests user/_grind user/_wc user/_zombie user/_logstress user/_forphan user/_dorphan user/_uptime
user/_find user/_memdump
nmeta 47 (boot, super, log blocks 31, inode blocks 13, bitmap blocks 1) blocks 1953 total 2000
ballocc: first 1039 blocks have been allocated
ballocc: write bitmap block at sector 46
qemu-system-riscv64 -machine virt -bios none -kernel kernel/kernel -m 128M -smp 3 -nographic -global virtio-mmio.force-legacy=false -drive file=fs.i
mg,if=none,format=raw,id=x0 -device virtio-blk-device,drive=x0,bus=virtio-mmio-bus.0

xv6 kernel is booting

hart 1 starting
hart 2 starting
init: starting sh
$ uptime 22
uptime 22
uptime: 38 ticks
$ |
```

Q2

```
dua_ansari@DESKTOP-4DF8P8: ~$ make qemu
ballocc: first 1039 blocks have been allocated
ballocc: write bitmap block at sector 46
qemu-system-riscv64 -machine virt -bios none -kernel kernel/kernel -m 128M -smp 3 -nographic -global virtio-mmio.force-legacy=false -drive file=fs.img,if=none,format=raw,id=x0 -device virtio
-blk-device,drive=x0,bus=virtio-mmio-bus.0

xv6 kernel is booting

hart 1 starting
hart 2 starting
init: starting sh
$ find . a*b
find . a*b
./zombie
$ find . test.*
find . test.*
./findtest.sh
./forktest
./usertests
$ find . c$
find . c$
./wc
$ find . .*
find . .*
./README
./findtest.sh
./sixfive.txt
./cat
./echo
./forktest
./grep
./init
./kill
./ln
./ls
./mkdir
./rm
./sh
./stressfs
./usertests
./grind
./wc
./zombie
./logstress
./forphan
./dorphan
./uptime
./find
./memdump
```

Q3 HISTORY AND TAB COMPLETION

```
dua_ansari@DESKTOP-4DF8P X + v
console 3 27 0
$ find . a*b
./zombie
$ history
1: ls
2: find . a*b
$ l
Completed to: ls
. 1 1 1024
.. 1 1 1024
README 2 2 2425
findtest.sh 2 3 91
sixfive.txt 2 4 19
cat 2 5 36512
echo 2 6 35368
forktest 2 7 17088
grep 2 8 39952
init 2 9 35816
kill 2 10 35288
ln 2 11 35104
ls 2 12 38648
mkdir 2 13 35352
rm 2 14 35336
sh 2 15 67032
stressfs 2 16 36216
usertests 2 17 186256
grind 2 18 51752
wc 2 19 27496
zombie 2 20 34688
logstress 2 21 37288
forphan 2 22 36112
dorphan 2 23 35560
uptime 2 24 34984
find 2 25 40704
memdump 2 26 36728
```

NO \$ WHEN PROCESSING COMMANDS FROM FILE

```
dua_ansari@DESKTOP-4DF8P X + v
echo first
echo second
sh mytest.sh

QEMU: Terminated
dua_ansari@DESKTOP-4DF8PF8:~/xv6-labs-2025$ make qemu
mv -f fs.img fs.img.bk
mkfs/mkfs fs.img README user/findtest.sh user/sixfive.txt user/_cat user/_echo user/_forktest user/_grep user/_init user/_kill user/_ln user/_ls use
r/_mkdir user/_rm user/_sh user/_stressfs user/_usertests user/_grind user/_wc user/_zombie user/_logstress user/_forphan user/_dorphan user/_uptime
user/_find user/_memdump
nmeta 47 (boot, super, log blocks 31, inode blocks 13, bitmap blocks 1) blocks 1953 total 2000
ballocc: first 1041 blocks have been allocated
ballocc: write bitmap block at sector 46
qemu-system-riscv64 -machine virt -bios none -kernel kernel/kernel -m 128M -smp 3 -nographic -global virtio-mmio.force-legacy=false -drive file=fs.i
mg,if=none,format=raw,id=x0 -device virtio-blk-device,drive=x0,bus=virtio-mmio-bus.0

xv6 kernel is booting

hart 2 starting
hart 1 starting
init: starting sh
$ cat > mytest.sh
echo first
echo second$ sh mytest.sh
first
second
```